01617 Introduction to Dynamical Systems (E25) - Assignment 1

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1) Problem 1 — Rescaling the Duffing Equation

We are given the Duffing equation:

$$M\frac{\mathrm{d}^2X}{\mathrm{d}T^2} + \alpha\frac{\mathrm{d}X}{\mathrm{d}T} = kX - \kappa X^3 + A\sin(\Omega T) \tag{1}$$

Into which we introduce the following linear transformations of the coordinates X and T:

$$T = k_t t \qquad X(T) = k_x x(t) \tag{2}$$

We consider the variable transformation of the first and second derivatives using the chain rule:

$$\begin{aligned}
\frac{\mathrm{d}X}{\mathrm{d}T} &= \frac{\mathrm{d}X}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}T} = \frac{k_x}{k_t}\frac{\mathrm{d}x}{\mathrm{d}t} \\
\frac{\mathrm{d}^2X}{\mathrm{d}T^2} &= \frac{\mathrm{d}}{\mathrm{d}T}\left[\frac{\mathrm{d}X}{\mathrm{d}T}\right] = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}T}\left[\frac{k_x}{k_t}\frac{\mathrm{d}x}{\mathrm{d}t}\right] = \frac{k_x}{k_t^2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2}
\end{aligned} \tag{3}$$

Plugging Equation 2 and Equation 3 into Equation 1, we obtain:

$$M\frac{k_x}{k_t^2}\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \alpha\frac{k_x}{k_t}\frac{\mathrm{d}x}{\mathrm{d}t} = kk_x x - \kappa k_x^3 x^3 + A\sin(\Omega k_t t) \tag{4}$$

1.a) Finding the Scaling Factors

We seek to find rescaling factors k_x, k_t such that we obtain the following form, correponding to normalisation of the force term $f(X) = kX - \kappa X^3$:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \beta\frac{\mathrm{d}x}{\mathrm{d}t} = x - x^3 + a\sin(\omega t) \tag{5}$$

We impose the condition that the coefficient on $\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ must be 1 by dividing Equation 4 through by $M\frac{k_x}{k_t^2}$:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\alpha k_t}{M}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{kk_t^2}{M}x - \frac{\kappa k_t^2 k_x^2}{M}x^3 + \frac{Ak_t^2}{Mk_x}\sin(\Omega k_t t) \tag{6}$$

Our remaining normalisation conditions imply:

$$\begin{aligned}
\frac{kk_t^2}{M} = 1 \quad &\Leftrightarrow \quad k_t = \pm\sqrt{\frac{M}{k}} \\
\frac{\kappa k_x^2 k_t^2}{M} = 1 \quad &\Leftrightarrow \quad k_x = \pm\sqrt{\frac{M}{\kappa k_t^2}}
\end{aligned} \tag{7}$$

Where we shall take the positive roots for both k_t and k_x for reasons discussed below in Section 1.b, such that:

$$k_t = \sqrt{\frac{M}{k}} \qquad \wedge \qquad k_x = \sqrt{\frac{k}{\kappa}} \tag{8}$$

Where we note that k_x appears to parameterise the relative strength of the linear and non-linear components of the force term $f(X)$, while k_t appears to parameterise the relative strength of the inertial term and force term $f(X)$.

Plugging in our scaling factors from Equation 8 into Equation 6, we obtain:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{\alpha}{\sqrt{kM}}\frac{\mathrm{d}x}{\mathrm{d}t} = x - x^3 + A\sqrt{\frac{\kappa}{k^3}}\sin\left(\Omega\sqrt{\frac{M}{k}}t\right) \tag{9}$$

Which when compared against Equation 5 reveals the rescales parameters as:

$$\beta = \frac{\alpha}{\sqrt{kM}} \qquad a = A\sqrt{\frac{\kappa}{k^3}} \qquad \omega = \Omega\sqrt{\frac{M}{k}} \tag{10}$$

1.b) Choice of Scaling Factors

There is some ambiguity in the question regarding choice of scaling factors k_x, k_t.

Firstly, one could have chosen a different *target equation* than Equation 5, for examply by choosing to normalise the driving term $A\sin(\Omega T)$ instead of the restoring force term $kX - \kappa X^3$.

This choice of scaling may benefit subsequent analysis of the system in some cases.

Alternatively, the question may refer to the ambiguity that arose in the choice of sign for k_x, k_t in Equation 7. We consider that the sign choice does not affect those terms where the scaling factors appear squared, but does affect the linear dampening term $\alpha\frac{\kappa_t}{M}\frac{\mathrm{d}x}{\mathrm{d}t}$ and the driving term $\frac{Ak_t^2}{Mk_x}\sin(\Omega k_t t)$.

We note that a choice of $k_t < 0$ would lead to a sign change in the dampening term, which would imply that the system is being driven rather than dampened. This is would be considered unphysical for most systems under consideration, so we may discard this choice.

Similarly, a choice of $k_x < 0$ would lead to a sign change in the driving term, which would imply a phase shift of π in the driving force. While this is not unphysical, it is a trivial phase shift that does not affect the dynamics of the system. For this reason, it is reasonable to discard this choice as well.

2) Problem 2 — Dynamical Analysis on Scalar First Order ODE

We are given the following first order ODE:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x) = (x+1)^2(1-x) \tag{11}$$

Where $x : \mathbb{R} \to \mathbb{R}$.

2.a) Sketch and Stationary Points

By inspection $f(x)$ has a double root at $x_0 = -1$ and a single root at $x_0 = 1$, which are thus the stationary points of the system where $f(x) = 0$.

Instead of computing the derivative of $f(x)$, we compute the sign of $f(x)$ in the intervals defined by the stationary points:

$$\begin{aligned} f(-2) &= +3 \\ f(-0) &= +1 \\ f(2) &= -9 \end{aligned} \tag{12}$$

We sketch the function $f(x)$ and its stationary points, as shown below in Figure 1

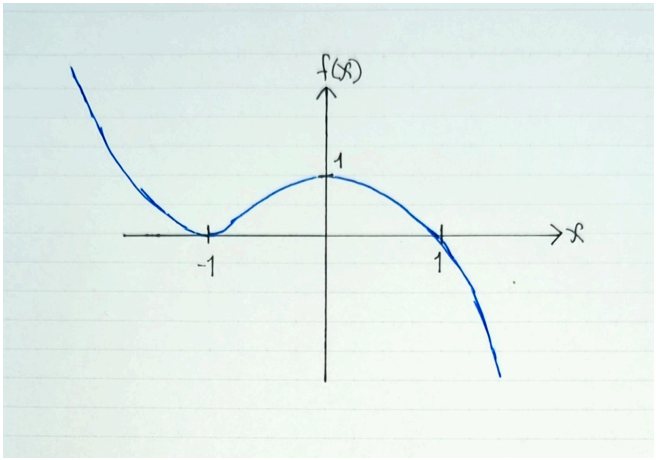


Figure 1: Sketch of $f(x)$ with stationary points at $x \in \{-1, 1\}$.

2.b) Solution Flow and Stability

We compute the derivative $f'(x_0)$:

$$f'(x) = -(x+1)(3x-1) \tag{13}$$

Which we evaluate at the stationary points:

$$f'(-1) = 0 \qquad \wedge \qquad f'(1) = -4 \tag{14}$$

We consider the single root at $x_0 = 1$, which has a negative derivative corresponding to a stable node.

Upon inspecting the stationary point at $x_0 = -1$, we find a vanishing derivative as expected from the double root, meaning we cannot determine stability from the linear approximation. Instead, we must consider the sign of the second derivative, $f''(x)$, in order to determine stability for this point:

$$\begin{aligned} f''(x) &= -6x - 2 \\ f''(-1) &= +4 \end{aligned} \tag{15}$$

To aide our understanding, we start at Ansatz $x(t) = x_0 + \eta(t)$ with temporal derivative $\frac{\mathrm{d}x}{\mathrm{d}t} = f(x) = \frac{\mathrm{d}\eta}{\mathrm{d}t}$ and perform a Taylor expansion of $f(x)$ around x_0, noting that we have already found $f(x_0) = f'(x_0) = 0$:

$$f(x) = \cancel{f(x_0)} + \cancel{f'(x_0)\eta} + \frac{1}{2}f''(x_0)\eta^2 + O(\eta^3) \tag{16}$$

Thus we find:

$$\begin{aligned} \frac{\mathrm{d}\eta}{\mathrm{d}t} &\approx \frac{1}{2}f''(x_0)\eta^2 \\ \left.\frac{\mathrm{d}\eta}{\mathrm{d}t}\right|_{x_0=-1} &\approx 2\eta^2 \end{aligned} \tag{17}$$

We may solve this ODE using separation of variables:

$$\begin{aligned} \frac{\mathrm{d}\eta}{\eta^2} &= 2\,\mathrm{d}t \\ \int \frac{1}{\eta^2}\,\mathrm{d}\eta &= \int 2\,\mathrm{d}t \\ \frac{1}{\eta} &= -2t + C \\ \eta(t) &= \frac{1}{-2t + C} \end{aligned} \tag{18}$$

We solve for the integration constant C using initial condition $\eta(0) = \eta_0$:

$$\eta(0) = \eta_0 = \frac{1}{-2(0) + C} = \frac{1}{C} \qquad \Rightarrow \qquad C = \frac{1}{\eta_0} \tag{19}$$

Thus obtaining:

$$\eta(t) = \frac{1}{\frac{1}{\eta_0} - 2t} = \frac{\eta_0}{1 - 2\eta_0 t} \tag{20}$$

This allows us to consider the stability of the point $x_0 = -1$ for time $t > 0$ based on the sign of the initial perturbation η_0:

$$\begin{aligned} \eta_0 > 0 \quad &\Rightarrow \quad \eta(t) \to +\infty \quad \text{as } t \to \frac{1}{(2\eta_0)_-} \quad \Rightarrow \quad \text{unstable} \\ \eta_0 < 0 \quad &\Rightarrow \quad \eta(t) \to 0_- \quad \text{as } t \to \infty \quad \Rightarrow \quad \text{stable from below} \end{aligned} \tag{21}$$

Thus we have *half-stability* at the point $x_0 = -1$, where the system is stable for perturbations from the left, but unstable for perturbations from the right.

We sketch the solution flow of the system in a phase line diagram using the information we have obtained as shown in Figure 2.

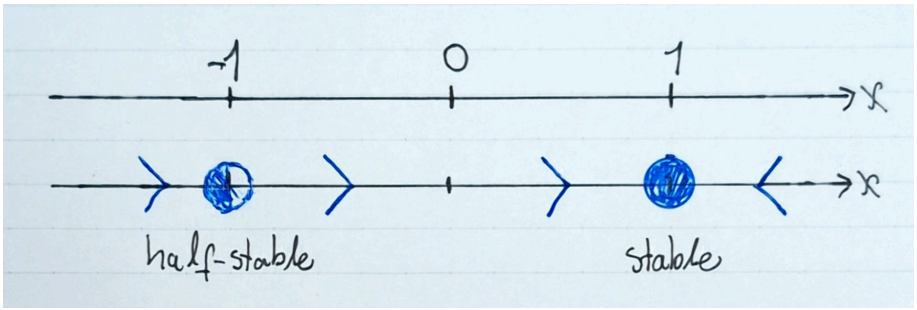


Figure 2: Solution flow of the system along the x-axis, showing stability of the stationary points.

As noted above, it is not possible to determine the stability of the point $x_0 = -1$ from the linear approximation alone, as $f'(-1) = 0$.

2.c) Sketching Solution Trajectories

We sketch the solutions $\phi_t(x_0)$ as a function of time $t > 0$ as shown in Figure 3.

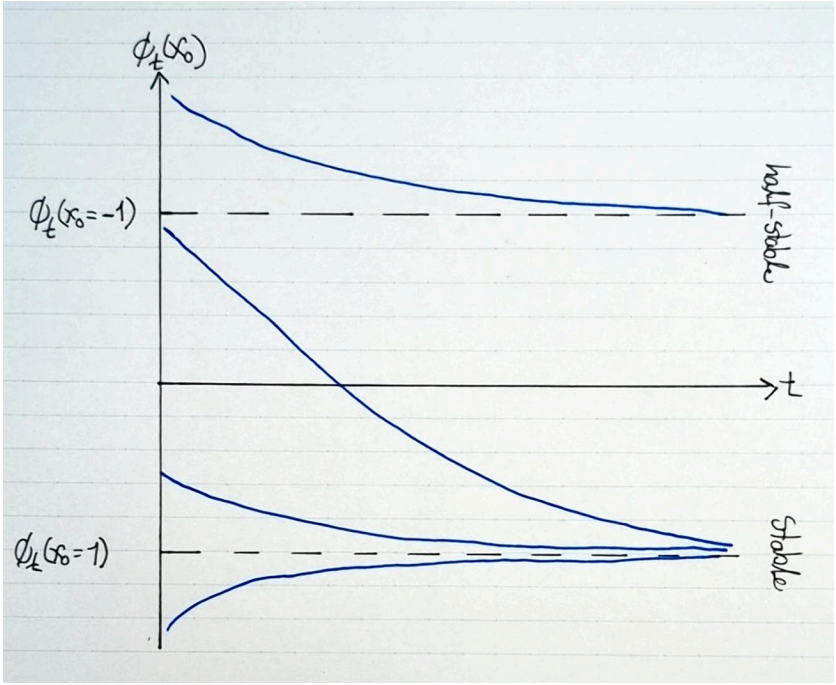


Figure 3: Sketch of solution trajectories $\phi_{t(x_0)}$ for various initial conditions x_0.

Where the half-stability of the point $x_0 = -1$ is evident from the trajectories approaching the point from below, but diverging away from it for initial conditions above the point.

2.d) Interval Solution

We will find a solution to Equation 11 using separation of variables and decomposition of rational functions.

First, we separate the variables to obtain the following integrals:

$$\int \frac{1}{(x+1)^2(1-x)}\,\mathrm{d}x = \int \mathrm{d}t \tag{22}$$

After which we decompose the spatial integrand, $\frac{1}{f(x)}$, into partial fractions:

$$\frac{1}{(x+1)^2(1-x)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{1-x} \tag{23}$$

We clear the denominators by multiplying through by $(x+1)^2(1-x)$ and collect in powers of x:

$$\begin{aligned} 1 &= A(x+1)(1-x) + B(1-x) + C(x+1)^2 \\ &= (C-A)x^2 + (2C-B)x + (A+B+C) \end{aligned} \tag{24}$$

From which we obtain the following system of equations:

$$\begin{aligned} C - A &= 0 \\ 2C - B &= 0 \\ A + B + C &= 1 \end{aligned} \tag{25}$$

Which we solve as:

$$A = \frac{1}{4} \qquad B = \frac{1}{2} \qquad C = \frac{1}{4} \tag{26}$$

We thus have the following integral to solve:

$$\int \frac{1}{4(x+1)} + \frac{1}{2(x+1)^2} + \frac{1}{4(1-x)}\,\mathrm{d}x = \int \mathrm{d}t \tag{27}$$

We do so by integrating the following terms and summing with the appropriate coefficients:

$$\begin{aligned}\int \frac{1}{x+1}\,\mathrm{d}x &= \ln|x+1| + C_1 \\ \int \frac{1}{(x+1)^2}\,\mathrm{d}x &= -\frac{1}{x+1} + C_2 \\ \int \frac{1}{1-x}\,\mathrm{d}x &= -\ln|1-x| + C_3 \\ \int \mathrm{d}t &= t + C_4\end{aligned} \tag{28}$$

Yielding solution:

$$\frac{1}{4}\ln\left(\frac{1+x}{1-x}\right) - \frac{1}{2}\frac{1}{1+x} = t - t_0 \tag{29}$$

Where we have combined the integration constants into a single constant t_0 on the right hand side and exploited the fact that the arguments of the logarithms are positive in the interval $x \in]-1;1[$ to drop the absolute values.

2.e) Limits of Interval Solution

We consider the solution given in Equation 29 in the limits $x \to -1_+$ and $x \to 1_-$.

We first consider the limit $x \to -1_+$, in which case the argument of the logarithm approaches zero from above, 0_+. We know that $\ln(0_+) \to -\infty$ and conclude that the logarithmic term goes to $-\infty$ in this limit. Similarly, by inspection, $\lim_{x\to -1_+}\left(\frac{1}{1+x}\right) = +\infty$ such that $-\frac{1}{2}\frac{1}{1+x} \to -\infty$. Thus, as both terms approach the same limit, we find that $t \to -\infty$ as $x \to -1_+$.

Secondly, we consider the limit $x \to 1_-$, in which case the argument of the logarithm approaches $+\infty$ such that $\lim_{x\to 1_-}\ln\left(\frac{1+x}{1-x}\right) = +\infty$. The rational term evaluates to $-\frac{1}{2}\frac{1}{1+1} = -\frac{1}{4}$ in the limit and is thus trivially dominated by the logarithmic term. As such, we conclude that $t \to +\infty$ as $x \to 1_-$.

3) Problem 3 — Dynamical Analysis on a Two-Dimensional System

We are given a system of differential equations:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{B}\mathbf{x} \qquad \text{where} \qquad \mathbf{B} = \begin{bmatrix}-1 & 2\\ 0 & -1\end{bmatrix}, \quad \mathbf{x}(0) = \mathbf{x}_0 \tag{30}$$

With $\mathbf{x} : \mathbb{R} \to \mathbb{R}^2$ and $\mathbf{x}_0 = (x_{01}, x_{02})^\mathsf{T}$.

3.a) Solution using Matrix Exponentials

We invoke the Fundamental Theorem of Linear Systems as given by [1, Chapter 1.4], which states that a system of the form Equation 30 has a unique solution given by:

$$\mathbf{x}(t) = e^{\mathbf{B}t}\mathbf{x}_0 \tag{31}$$

As such, our problem has been reduced to finding the matrix exponential $e^{\mathbf{B}t}$.

3.a.1) Solution using Coordinate Transformation

While this problem is easily solved using [1, Chapter 1.4, Corollary 4], we will instead opt for the idiomatic approach using a change of basis to cast the problem into its Jordan normal form. We do this by first determining the eigenvalues of $\mathbf{B}$ to be $\lambda_1 = \lambda_2 = -1$ using the known result that the eigenvalues of an upper triangular matrix are given by its diagonal elements.

We note that the eigenvalues are equal and that we thus have an *algebraic multiplicity* of 2, but we only have a *geometric multiplicity* of 1. As such, we need to find a *generalized eigenvector* in addition to the *regular eigenvector*.

The eigenvectors, $\mathbf{v}$ are found by solving the familiar equation:

$$(\mathbf{B} - \lambda\mathbf{I})\mathbf{v} = 0 \tag{32}$$

Which trivially yields the eigenvector:

$$\mathbf{v}_1 = (1, 0)^\mathsf{T} \tag{33}$$

As dictated by [1, Chapter 1.7], we may find the generalized eigenvector $\mathbf{v}_2$ which satisfies:

$$(\mathbf{B} - \lambda\mathbf{I})^k\mathbf{v} = 0 \tag{34}$$

Where k is the *degree of nilpotency* of the matrix $(\mathbf{B} - \lambda\mathbf{I})$, which we find as:

$$\begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad \Rightarrow \qquad k = 2 \tag{35}$$

We obtain $\mathbf{v}_2$ using the Jordan chain conditions, which can be used to recursively find generalized eigenvectors [2]:

$$(\mathbf{B} - \lambda\mathbf{I})\mathbf{v}_k = \mathbf{v}_{k-1} \tag{36}$$

Thus, we find $\mathbf{v}_2$ by solving:

$$\begin{aligned} (\mathbf{B} - \lambda\mathbf{I})\mathbf{v}_2 &= \mathbf{v}_1 \\ \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}\mathbf{v}_2 &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{aligned} \tag{37}$$

From which we find a general solution, from which we may pick $c = 0$ to obtain a particular solution:

$$\mathbf{v}_2 = (c, 1/2)^\mathsf{T} = (0, 1/2)^\mathsf{T} \tag{38}$$

As explained in [1, Chapter 1.6], we may now construct an invertible matrix $\mathbf{P}$ using our eigenvectors as columns:

$$\mathbf{P} = [\mathbf{v}_1 \;\; \mathbf{v}_2] = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \tag{39}$$

Which transforms $\mathbf{B}$ into its Jordan normal form $\mathbf{C}$:

$$\mathbf{C} = \mathbf{P}^{-1}\mathbf{B}\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \tag{40}$$

Which we confirm as a Jordan normal form matching *Case II* of [1, Chapter 1.5], allowing us to characterise the system as an *improper stable node* given $\lambda_1 = \lambda_2 < 0$.

We may now leverage the properties described in [1, Chapter 1.3] to find:

$$e^{\mathbf{B}t} = \mathbf{P}e^{\mathbf{C}t}\mathbf{P}^{-1} \tag{41}$$

Where in the case of $\mathbf{C} = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$ we have the known result that:

$$e^{\mathbf{C}t} = e^{\lambda t}\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \tag{42}$$

Which yield the solution:

$$e^{\mathbf{B}t} = \mathbf{P}e^{\mathbf{C}t}\mathbf{P}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}e^{-t}\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = e^{-t}\begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix} \tag{43}$$

Such that the solution to Equation 30:

$$\mathbf{x}(t) = e^{-t}\begin{bmatrix}1 & 2t\\ 0 & 1\end{bmatrix}\mathbf{x}_0 \tag{44}$$

3.a.2) Alternative Approach

The treatment above was rather cumbersome, but was undertaken to practice the method of transforming to Jordan normal form and back.

Given the rather simple form of $\mathbf{B}$, we may have instead opted for a more direct approach to finding the matrix exponential.

As before, we will utilise the Fundamental Theorem of Linear Systems as given by [1, Chapter 1.4], such that the problem has been reduced to finding the matrix exponential $e^{\mathbf{B}t}$.

We may decompose $\mathbf{e^{Bt}}$ as follows:

$$\begin{aligned}\mathbf{e^{Bt}} = \exp\left(\begin{bmatrix}-t & 2t\\ 0 & -t\end{bmatrix}\right) &= \exp\left(\begin{bmatrix}-t & 0\\ 0 & -t\end{bmatrix} + \begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right)\\ &= \exp\left(-t\mathbf{I} + \begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right)\end{aligned} \tag{45}$$

We seek to prove that $-t\mathbf{I}$ and $\begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}$ commute, which is trivial upon noting that $t\mathbf{I}$ is a *scalar matrix* and thus commutes with all matrices of the same dimension. This enables the decomposition of the exponential of a sum into a product of exponentials as per [1, Chapter 1.3, Proposition 2]:

$$\begin{aligned}\mathbf{e^{Bt}} &= \exp\left(-t\mathbf{I} + \begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right)\\ &= e^{-t\mathbf{I}}\exp\left(\begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right)\\ &= e^{-t}\mathbf{I}\exp\left(\begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right)\end{aligned} \tag{46}$$

We can now evaluate $\exp\left(\begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right)$ using the definition of the matrix exponential as a power series:

$$e^{\mathbf{S}} = \lim_{n\to\infty}\sum_{k=0}^{n}\frac{\mathbf{S}^k}{k!} \tag{47}$$

Where we note that $\mathbf{S} = \begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}$ is *nilpotent* of degree 2 as $\mathbf{S}^2 = \mathbf{0}$, such that the power series truncates after the first order term:

$$\exp\left(\begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix}\right) = \mathbf{I} + \begin{bmatrix}0 & 2t\\ 0 & 0\end{bmatrix} + \mathbf{0} = \begin{bmatrix}1 & 2t\\ 0 & 1\end{bmatrix} \tag{48}$$

Combining these results, we find:

$$\mathbf{e^{Bt}} = e^{-t}\mathbf{I}\begin{bmatrix}1 & 2t\\ 0 & 1\end{bmatrix} = e^{-t}\begin{bmatrix}1 & 2t\\ 0 & 1\end{bmatrix} \tag{49}$$

Which is consistent with the treatment in Section 3.a.1, yielding the same solution to Equation 30:

$$\mathbf{x}(t) = e^{-t}\begin{bmatrix}1 & 2t\\ 0 & 1\end{bmatrix}\mathbf{x}_0 \tag{50}$$

3.b) Phase Portrait and Stability

Our solutions to Section 3.a can aid us in sketching the phase portrait of the system.

The degenerate eigenvalue $\lambda_1 = \lambda_2 = -1 < 0$ indicates that the system is *stable* at the stationary point $\mathbf{x} = (0, 0)^\mathsf{T}$.

As such, for all points (except the trivial case of the origin), we expect the trajectories to point towards the origin as $t \to \infty$. From [1, Chapter 1.5, Case II], we know that the trajectories will be tangent to the eigenvector $\mathbf{v}_1 = (1, 0)^\mathsf{T}$ as they approach the origin.

Lastly, we consider the coordinate transformation matrix $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ from Section 3.a.1, which indicates that the y-coordinate is scaled by a factor of $\frac{1}{2}$ in the eigenbasis. This means that the trajectories will be compressed in the y-direction relative to the x-direction, which will be largely inconsequential for the sketch of the phase portrait.

With this, we are able to sketch the phase portrait of the system as shown in Figure 4.

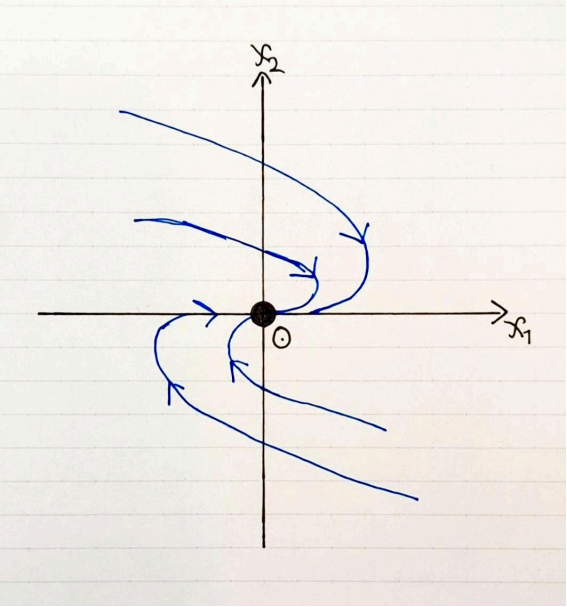


Figure 4: Phase portrait of the system showing trajectories approaching the stable, improper node at the origin.

3.c) Invariant Sets and Subspaces

We understand an invariant set to be a set such that if the system starts in the set, it remains in the set for all time $t > 0$. We recall from [1, Chapter 1.9] that we may write the phase space as the outer product of the stable, unstable, and center subspaces given by E^s, E^u, and E^c respectively:

$$\mathbb{R}^2 = E^s \oplus E^u \oplus E^c \tag{51}$$

Where the subspaces may be found using the spans of the eigenvectors corresponding to the eigenvalues with negative, positive, and zero real parts respectively, for $i \in \{1, 2\}$:

$$\begin{aligned} E^s &= \text{Span}\{\mathbf{v}_i \mid \Re(\lambda_i) < 0\} \\ E^u &= \text{Span}\{\mathbf{v}_i \mid \Re(\lambda_i) > 0\} \\ E^c &= \text{Span}\{\mathbf{v}_i \mid \Re(\lambda_i) = 0\} \end{aligned} \tag{52}$$

As such, we find using the eigenvectors $\lambda_1 = \lambda_2 = -1 < 0$:

$$E^s = \mathbb{R}^2 \qquad E^u = \{0\} \qquad E^c = \{0\} \tag{53}$$

We may subdivide the stable subspace, which in this case is the entire phase space, smaller invariant sets as well. Trivially, the stationary point at the origin is an invariant set:

$$\{(0, 0)^\mathsf{T}\} \subset E^s \tag{54}$$

Additionally, as illustrated by the sketch in Figure 4 and by inspection of Equation 30, we note that the first axis itself is an invariant set, as any point on the axis will remain on the axis for all time $t > 0$:

$$\left\{(x_1, 0)^\mathsf{T} \mid x_1 \in \mathbb{R}\right\} \subset E^s \tag{55}$$

Trivially, the entire phase space is of course also an invariant set.

In conclusion, we find the following invariant sets:

$$\left\{(0,0)^\mathsf{T}\right\} \subset \left\{(x_1, 0)^\mathsf{T} \mid x_1 \in \mathbb{R}\right\} \subset \mathbb{R}^2 \tag{56}$$

Which are all stable invariant sets.

Bibliography

[1] L. Perko, *Differential Equations and Dynamical Systems*, vol. 7. in Texts in Applied Mathematics, vol. 7. New York, NY: Springer New York, 2001. doi: 10.1007/978-1-4613-0003-8.

[2] C. Soerensen Mads Peter; Henriksen, "01617 Introduction to Dynamical Systems Week 3 - Course Lecture Notes," 2025.